FIN 522A: Lecture 1

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- Brendan Griffin holds an appointment at Duke University's Pratt School of Engineering, where he is focused on finance and valuation. He has spent his career building, funding, and selling innovative companies and technologies, and is a former investment banker and senior corporate executive with extensive experience executing large, complex transactions in the realms of the global capital markets, M&A, and corporate restructuring. He is experienced in complex valuation, financial modeling, due diligence, and advanced corporate finance and governance.
- Brendan has closed over 100 deals worth more than $15 billion in public and private financings and strategic transactions, including IPOs and complex cross-border deals, across North America, Europe, and Asia. He previously served as Chief Financial Officer of a clinical-stage biotechnology company acquired by Astellas, a large Japanese multinational company.
- Brendan holds an MBA from Carnegie Mellon University's Tepper School of Business, a Master of Public Policy Management from Carnegie Mellon's Heinz School, and a B.S. from Elon University.

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Course Introduction to FIN 522A

- **What this course is not:**
  - a pure corporate finance course
- **What this course is:**
  - an applied asset pricing course
  - focused on how markets price cash flows and risk across asset classes
- **Assumes:**
  - You have a solid foundation in accounting and corporate finance
- **By the end of this course:**
  - you will have a strong conceptual and technical foundation in asset pricing
  - you will be able to analyze and compare pricing across different asset classes

The Corporate Finance Starting Point

- Standard DCF framework:

$$PV = \sum \frac{CF_t}{(1 + WACC)^t}$$

- Cost of equity, WACC, hurdle rates
- Where this works well:
  - Internal projects
  - Capital budgeting
  - Long-run planning

What Standard DCF Assumes About Discounting

- A single discount rate (WACC or cost of equity) is applied to:
  - all maturities
  - all cash-flow timing
- Risk is summarized into one number, rather than maturity-specific pricing
- Discount rate is modeled, not directly observed
- Complexity can be integrated with cash flows being:
  - probability-weighted
  - reinvested into capex or growth
  - scenario-based
- But discounting itself remains flat across time (at a standard level at least)

When Is Standard DCF Appropriate?

- Standard DCF works best when:
  - cash flows are uncertain and endogenous (e.g. equities)
  - discount rates are not observable
  - reinvestment and growth decisions matter
- Standard DCF is limited when:
  - cash flows are contractual (e.g. bonds)
  - prices exist across maturities
  - markets already imply discount factors

In traded fixed-income markets, discount rates are inferred from market prices, not assumed.

Market Pricing

- Prices are determined through trading, not models
- Market prices reflect:
  - expectations about future cash flows
  - liquidity conditions
  - supply and demand imbalances
  - hedging needs and risk transfer
- These forces are embedded directly in observed prices
- In traded fixed-income markets discount rates are inferred from prices, not assumed

Market Pricing & Fixed Income

- Markets price cash flows using observable traded prices
- Fixed-income securities:
  - have contractual cash flows
  - trade across many maturities
  - allow discounting to be inferred directly from prices
- Fixed income our natural starting point for market-based pricing:
  - discount factors
  - spot rates
  - forward rates
- We will revisit the more complex valuation frameworks where markets do not directly reveal discount rates later (e.g., equities and private markets)

Discount Factors from Market Prices

- A discount factor is the market price today of $1 received at a future date
- Denoted: $DF(n)$
- Also called “Present Value Factor” or PV Factor
- In traded fixed-income markets:
  - prices are observed
  - discount factors are implied directly from those prices
- Interest rates and yields are derived summaries

Spot Rates and Discount Factors

- Spot rates summarize the total return over n periods
- Let z_n denote the n -period spot rate
- Defined by the discount factor:

$$DF(n) = \frac{1}{(1 + z_n)^n}$$

- Spot rates summarize the time value of money
- They do not directly price individual cash flows

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Spot Rates as Compounded Short Rates

- A short rate for a given time interval refers to the interest rate for that interval available at different points in time
- Consider a certain world, that is a world where investors know the path of future interest rates:
  - Let r_n denote the one-period short rate from time $n-1$ to n
  - The n-period spot rate satisfies:

$$(1+z_n)^n=(1+r_1)(1+r_2)\cdots(1+r_n)$$

  - Spot rates summarize a sequence of one-period rates

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Future Short Rates in Certain World…

- Assume future short rates are known with certainty today and there is no uncertainty or risk premia
- Consider two investment strategies ending at time n:
  - lock in an n-period spot rate today
  - invest for $n-1$ periods, then lock in a rate for period $n-1 \rightarrow n$
- No-arbitrage requires both strategies to have the same value today
- In a certain world, this condition defines the future short rate:

$$(1+z_n)^n = (1+z_{n-1})^{n-1}(1+r_n)$$

Or

$$1 + r_n = \frac{(1+z_n)^n}{(1+z_{n-1})^{n-1}}$$

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…But Forward Rate in an Uncertain World

- However, in our world future interest rates are unknown, so what is this rate?
- What we can infer in this matter is the forward rate:
  - let z_n denote the spot rate to time n, and f_n the one-period forward rate from n−1 to n:

$$1 + f_n = \frac{(1 + z_n)^n}{(1 + z_{n-1})^{n-1}}$$

- So does $f_n = \mathbb{E}[r_n]$?

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Theories of the Term Structure

Three classic theories:

1. Expectations Theory
2. Market Segmentation Theory
3. Liquidity Preference Theory

Expectations Theory

- Assumes:
  - no term or liquidity premia
  - investors are risk-neutral across maturities
- Forward rates equal expected future short rates:

$$f_n = \mathbb{E}[r_n]$$

- Yield curve reflects pure expectations of future policy rates
- Implication:
  - Upward-sloping curve ⇒ expected rate hikes
  - Inverted curve ⇒ expected rate cuts

Limits of Expectations Theory

- Forward rates are biased predictors of future short rates
- Cannot explain:
  - persistent upward-sloping yield curves
  - time-varying term premia
- Ignores risk, liquidity, and balance-sheet constraints
- Conclusion: Expectations alone do not price the yield curve, so

$f_n = \mathbb{E}[r_n]$ does not hold

Market Segmentation Theory

- Investors have preferred maturities:
  - money market funds → short
  - pensions / insurers → long
- Limited substitution across maturities
- Yield curve reflects:
  - supply and demand by maturity
  - institutional constraints
- Limitation: Explains shape, but lacks a pricing mechanism

Liquidity Preference Theory

- Investors prefer shorter maturities
- Longer maturities require compensation for:
  - interest-rate risk
  - liquidity risk
  - funding uncertainty
- Forward rates satisfy:

  $f_n = \mathbb{E}[r_n]$ + liquidity premium (can be positive or negative)
- Liquidity Preference Theory is most appealing

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Interpreting the Yield Curve

- Upward-sloping curve:
  - positive term premium
  - not necessarily expected tightening
- Flattening curve:
  - declining premia or rising demand for duration
- Inverted curve:
  - expectations of easing and/or strong demand for safe long assets

Session 1 Summary

- Market discounting is inferred from prices
- Discount factors are the fundamental objects
- Spot, short, and forward rates are linked but distinct
- Forward rates reflect expectations plus premia
- Liquidity preference provides the most complete framework

Looking Ahead

- STRIPS and bond stripping
- Bootstrapping discount factors
- Constructing yield curves